

Human-Computer Interaction: Usability Design & Evaluation Studies

Comprehensive Notes based on Usability.gov and Interaction Design (Ch.
15)

Detailed Compilation

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Contents

1	Part I: Usability in Design	2
1.1	1. Core Objectives of Usability	2
1.2	2. Setting Measurable Usability Goals	2
	1.2.1 Categories of Usability Goals	2
	1.2.2 Levels of Measurement	3
1.3	3. Meeting User Needs	3
1.4	4. Heuristic Evaluation	3
	1.4.1 Jakob Nielsen's 10 Heuristics	3
	1.4.2 Gerhardt-Powals Heuristics (Cognitive Engineering)	4
2	Part II: Evaluation Studies	5
2.1	1. Overview of Evaluation Methods	5
2.2	2. Usability Testing (Lab-Based)	5
2.3	3. Remote Testing	5
2.4	4. Case Studies	6
	2.4.1 Case 1: iPad Usability (Budiou & Nielsen, 2010)	6
	2.4.2 Case 2: Remote VR Testing (Sanni Siltanen et al., 2021)	6
	2.4.3 Case 3: Painpad (In-the-Wild Study)	6
2.5	5. Experiments (Research Focus)	7
	2.5.1 Key Components	7
	2.5.2 Experimental Designs	7
2.6	6. In-the-Wild Studies	7

1 Part I: Usability in Design

Based on: Usability.gov slides

1.1 1. Core Objectives of Usability

To create a successful design, one must align three competing interests:

1. **Organization's Objectives:** What the business wants to achieve (e.g., sales, lead generation).
2. **User's Needs:** What the visitor wants to achieve (e.g., finding information, completing a task quickly).
3. **Web Guidelines:** Compliance with standards (e.g., W3C, Accessibility).

1.2 2. Setting Measurable Usability Goals

Usability goals must be defined during ****Kickoff Meetings****. A goal is only useful if it is measurable.

1.2.1 Categories of Usability Goals

- **Time:** How long does it take to complete a task?
 - Time to locate the application/information.
 - Time to recover from an error.
- **Accuracy:** How many errors occur?
 - Number of unproductive searches.
 - Misinterpretation of information.
- **Overall Success:** Can the user complete the task without help?
 - *Metric Example:* A user should return to their task within 2 minutes after consulting the help menu.
- **Satisfaction:** Subjective happiness of the user (measured via Likert scales).

Critical Insight: Performance vs. Satisfaction

Slides note that users often give high satisfaction ratings (to be polite or because they blame themselves) even when they fail tasks. Therefore, **Performance measures (Time/Accuracy) are more reliable than Satisfaction ratings.**

1.2.2 Levels of Measurement

1. **Website Level:** High-level goals.
 - *Example:* "95% of travelers will be able to make their own airline reservations."
2. **Scenario Level:** Based on specific workflows involving multiple pages.
 - *Example:* "90% of users will find a specific thyroid cancer article within 3 minutes."
3. **Page Level:** Micro-interactions on a specific screen.
 - *Example:* "The page will download in 5 seconds or less on broadband."

1.3 3. Meeting User Needs

To ensure the site fulfills user goals, designers must research who is visiting and why.

- **Feedback Analysis:** Review emails and support call logs.
- **Log Analysis:** Evaluate web logs and search queries (what are people looking for that they can't find?).
- **Surveys & Tests:** Conduct online surveys or formal usability tests.

1.4 4. Heuristic Evaluation

A usability inspection method where a small set of evaluators (1–3 experts) examine the interface against established principles (heuristics).

1.4.1 Jakob Nielsen's 10 Heuristics

1. **Visibility of system status:** The user should always know what is happening (e.g., loading spinners).
2. **Match between system and real world:** Use familiar language and metaphors (e.g., trash can icon).
3. **User control and freedom:** Support "Undo" and "Redo."
4. **Consistency and standards:** Follow platform conventions.
5. **Error prevention:** Design to prevent slips before they happen (e.g., confirmation dialogs).
6. **Recognition rather than recall:** Minimize cognitive load; don't make users remember data from one screen to the next.
7. **Flexibility and efficiency of use:** Accelerators for experts (keyboard shortcuts).
8. **Aesthetic and minimalist design:** Remove irrelevant information.
9. **Help users recognize, diagnose, and recover from errors:** Plain language error messages.
10. **Help and documentation:** Easy to search and focused on tasks.

1.4.2 Gerhardt-Powals Heuristics (Cognitive Engineering)

These focus more on data processing and cognitive load:

- **Automate unwanted workload:** Do calculations for the user.
- **Reduce uncertainty:** Display data clearly.
- **Fuse data:** Combine lower-level data into high-level summaries.
- **Judicious Redundancy:** Provide multiple ways to perceive data (e.g., color coding + text labels).

2 Part II: Evaluation Studies

Based on: Interaction Design, Chapter 15: From Controlled to Natural Settings

2.1 1. Overview of Evaluation Methods

Evaluation is crucial to determine if a system is usable and to discover knowledge.

- **Usability Testing:** Controlled settings, focuses on performance improvement.
- **Experiments:** Scientific research, focuses on testing hypotheses and discovering new knowledge.
- **In-the-Wild Studies:** Natural settings, focuses on how technology is used in real life.

2.2 2. Usability Testing (Lab-Based)

Definition: Testing how people perform on products in controlled settings (labs).

- **Setup:** Often uses a one-way mirror. Designers watch from an observation room while a facilitator interacts with the user.
- **Data Collected:**
 - *Quantitative:* Time to complete, number of errors, navigation path, recovery time.
 - *Qualitative:* Video recording, keystroke logging, user satisfaction surveys.
- **Sample Size:** typically **5–10 participants** is sufficient to identify serious problems.

Real-World Example: Modern Equipment Trends

Equipment is shrinking. Heavy tripods and head-mounted racks are being replaced by:

- **Tobii Glasses:** Wearable eye-tracking that looks like normal eyewear.
- **Mobile Phones:** High-quality video/audio recording eliminates the need for bulky cameras.

2.3 3. Remote Testing

Testing where the evaluator and participant are separated in space (and sometimes time).

- **Types:**
 - *Synchronous:* Live via Zoom/Teams.
 - *Asynchronous:* User completes tasks on their own time; software records the screen.

- **Drivers:** The COVID-19 pandemic accelerated the adoption of remote testing.
- **Pros:** Can test more people globally; participants are in their natural environment; better for participants with disabilities.
- **Cons:** Cannot fix technical issues (hardware failures); lack of personal connection.

2.4 4. Case Studies

2.4.1 Case 1: iPad Usability (Budi & Nielsen, 2010)

- **Context:** Conducted secretly before iPad launch to help 3rd party developers.
- **Method:** Lab-based, rapid testing in two cities (Fremont, Chicago). 7 participants with iPhone experience.
- **Procedure:**
 1. Informed consent (confidentiality was crucial).
 2. Exploratory phase (free use).
 3. Specific tasks (e.g., "Find a recipe for apple pie on Epicurious," "Buy an iPad on eBay").
- **Findings:** Users struggled with "tap" areas (affordance issues), got lost in apps, and found web access difficult.
- **Limitation:** It was a short-term lab study; it did not capture how people fit the iPad into their daily lives over time.

2.4.2 Case 2: Remote VR Testing (Sanni Siltanen et al., 2021)

- **Context:** Testing a collaborative Extended Reality (XR) platform during the COVID-19 pandemic.
- **Challenge:** How to test hardware remotely when sanitization is required?
- **Setup:** "DesignSpace" environment tested across 8 countries (Finland, China, USA, etc.).
- **Method:** Hybrid. Some tests in-person (with strict disinfecting), others fully remote. Used Head-Mounted Displays (HMDs).

2.4.3 Case 3: Painpad (In-the-Wild Study)

- **Device:** A tangible keypad for patients to log pain levels after surgery.
- **Setting:** Two UK hospitals. Natural environment (post-surgery recovery).
- **Participants:** 54 patients (Mean age 64.6).
- **Method:** Device prompted patients every 2 hours to log pain. Data was compared against nurse charts.

- **Findings:**

- Satisfaction was high (4.63/5 Likert).
- Patients logged *more* data points than nurses did.
- Compliance was mixed (some didn't hear the prompt).
- **Privacy:** A major concern in a shared hospital ward.

2.5 5. Experiments (Research Focus)

Unlike usability testing (which improves a product), experiments aim to validate a hypothesis and discover new scientific knowledge.

2.5.1 Key Components

- **Hypothesis:** A prediction of a relationship.
- **Independent Variable (IV):** Manipulated by the researcher (e.g., Button Size).
- **Dependent Variable (DV):** Measured outcome (e.g., Time to Click).
- **Control:** Keeping other variables constant.

2.5.2 Experimental Designs

Design Type	Description	Pros/Cons
Between-Subjects	Different participants for each condition.	Pros: No learning effect. Cons: High individual variance; requires many participants.
Within-Subjects	Same participants do all conditions.	Pros: Fewer participants needed; eliminates individual differences. Cons: Learning/Order effects (requires counterbalancing).
Matched (Pair-wise)	Participants matched by criteria (e.g., expertise).	Reduces individual differences but hard to match perfectly.

Table 1: Comparison of Experimental Designs

2.6 6. In-the-Wild Studies

These studies deploy prototypes in natural settings (homes, offices, streets).

- **Goal:** Understand how technology impacts behavior and how users appropriate technology in unexpected ways.
- **Long-term Studies:**

- Used for complex products (e.g., visualization tools) that take weeks to learn.
- **Method:** Deployment for 2–4 weeks.
- **Data Collection:** Diaries, automatic usage logging, interviews.